

Contents

Preface to the Second Edition	vii
Preface to the First Edition	ix
1 Introduction	1-1
1.1 What Is Feedback?	1-1
1.2 What is Feedforward?	1-4
1.3 What Is Control?	1-5
1.4 Uses of Feedback and Control	1-6
1.5 Feedback Properties	1-13
1.6 Simple Forms of Feedback	1-18
1.7 Combining Feedback with Logic	1-20
1.8 Control System Architectures	1-24
1.9 Further Reading	1-27
Exercises	1-28
2 Feedback Principles	2-1
2.1 Nonlinear Static Models	2-1
2.2 Linear Dynamical Models	2-5
2.3 Using Feedback to Attenuate Disturbances	2-12
2.4 Using Feedback to Track Reference Signals	2-17
2.5 Using Feedback to Provide Robustness	2-20
2.6 Positive Feedback	2-23
2.7 Further Reading	2-28
Exercises	2-28
3 System Modeling	3-1
3.1 Modeling Concepts	3-1
3.2 State Space Models	3-10
3.3 Modeling Methodology	3-22
3.4 Modeling Examples	3-30
3.5 Further Reading	3-44
Exercises	3-44

4	Examples	4-1
4.1	Cruise Control	4-1
4.2	Bicycle Dynamics	4-5
4.3	Operational Amplifier Circuits	4-8
4.4	Computing Systems and Networks	4-11
4.5	Atomic Force Microscopy	4-17
4.6	Drug Administration	4-21
4.7	Population Dynamics	4-26
	Exercises	4-28
5	Dynamic Behavior	5-1
5.1	Solving Differential Equations	5-1
5.2	Qualitative Analysis	5-4
5.3	Stability	5-8
5.4	Lyapunov Stability Analysis	5-18
5.5	Parametric and Nonlocal Behavior	5-29
5.6	Further Reading	5-34
	Exercises	5-35
6	Linear Systems	6-1
6.1	Basic Definitions	6-1
6.2	The Matrix Exponential	6-6
6.3	Input/Output Response	6-15
6.4	Linearization	6-28
6.5	Further Reading	6-34
	Exercises	6-35
7	State Feedback	7-1
7.1	Reachability	7-1
7.2	Stabilization by State Feedback	7-9
7.3	Design Considerations	7-17
7.4	Integral Action	7-24
7.5	Linear Quadratic Regulators	7-28
7.6	Further Reading	7-33
	Exercises	7-33
8	Output Feedback	8-1
8.1	Observability	8-1
8.2	State Estimation	8-6
8.3	Control Using Estimated State	8-11
8.4	Kalman Filtering	8-17
8.5	State Space Controller Design	8-23
8.6	Further Reading	8-32
	Exercises	8-32

9	Transfer Functions	9-1
9.1	Frequency Domain Modeling	9-1
9.2	Determining the Transfer Function	9-3
9.3	Laplace Transforms	9-14
9.4	Block Diagrams and Transfer Functions	9-17
9.5	Zero Frequency Gain, Poles, and Zeros	9-23
9.6	The Bode Plot	9-29
9.7	Further Reading	9-40
	Exercises	9-41
10	Frequency Domain Analysis	10-1
10.1	The Loop Transfer Function	10-1
10.2	The Nyquist Criterion	10-4
10.3	Stability Margins	10-14
10.4	Bode's Relations and Minimum Phase Systems	10-18
10.5	Generalized Notions of Gain and Phase	10-22
10.6	Further Reading	10-27
	Exercises	10-28
11	PID Control	11-1
11.1	Basic Control Functions	11-1
11.2	Simple Controllers for Complex Systems	11-7
11.3	PID Tuning	11-11
11.4	Integral Windup	11-15
11.5	Implementation	11-19
11.6	Further Reading	11-24
	Exercises	11-24
12	Frequency Domain Design	12-1
12.1	Sensitivity Functions	12-1
12.2	Performance Specifications	12-6
12.3	Feedback Design via Loop Shaping	12-13
12.4	Feedforward Design	12-18
12.5	The Root Locus Method	12-24
12.6	Design Example	12-27
12.7	Further Reading	12-30
	Exercises	12-31
13	Robust Performance	13-1
13.1	Modeling Uncertainty	13-1
13.2	Stability in the Presence of Uncertainty	13-9
13.3	Performance in the Presence of Uncertainty	13-15
13.4	Design for Robust Performance	13-19
13.5	Further Reading	13-28
	Exercises	13-28

14 Fundamental Limits	14-1
14.1 System Design Considerations	14-1
14.2 Bode's Integral Formula	14-5
14.3 Gain Crossover Frequency Inequality	14-10
14.4 The Maximum Modulus Principle	14-15
14.5 Robust Pole Placement	14-20
14.6 Nonlinear Effects	14-26
14.7 Further Reading	14-30
Exercises	14-30
15 Architecture and System Design	15-1
15.1 Introduction	15-1
15.2 System and Control Design	15-2
15.3 Top-Down Architectures	15-6
15.4 Bottom-Up Architectures	15-17
15.5 Interaction	15-24
15.6 Adaptation and Learning	15-29
15.7 Control Design in Common Application Fields	15-37
15.8 Further Reading	15-42
Bibliography	B-1
Index	I-1